A PRODUCT OF FIELDS IS ABSOLUTELY FLAT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A ring in which every finitely generated ideal is generated by an idempotent is absolutely flat, characterized by every prime being maximal and the local rings being fields. invoked at proof lines 30298-30299. Supplies the criterion invoked at the end.

Source reference. Original lemma 092F: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30250–30250. Cited at proof lines 30298-30299 for the concluding criterion; the conclusion is used inside the counted span.

Source reference. Original lemma 04MG: sources/stacks-chow/tags.txt, source lines 6042–6042. Cited at proof lines 30298-30299 together with the criterion; the conclusion is used inside the counted span.

Lemma 0.2. *A product of fields is an absolutely flat ring.*

Proof. Let K_i be a family of fields. If $f = (f_i) \in \prod K_i$, then the ideal generated by f is the same as the ideal generated by the idempotent $e = (e_i)$ with $e_i = 0, 1$ according to whether f_i is 0 or not. Thus $D(f) = D(e)$ is open and closed and we conclude by Lemma 092F and Algebra, Lemma 04MG. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.